विशाल जाधव सरांचे

VJTech Academy

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"UNIT-III Inheritance"

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Program: Single Inheritance

```
import java.util.*;
class Student
{
	int rollno;
	String name;
	void get_stud_info()
	{
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter Student Roll No:");
		rollno=sc.nextInt();
		System.out.println("Enter Student Name:");
		name=sc.next();
	}
	void disp_stud_info()
	{
		System.out.println("Roll No:"+rollno);
		System.out.println("Student Name:"+name);
	}
}
class Test extends Student
{
	int marks1,marks2;
	void get_marks()
	{
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter Class Test-1 Marks:");
		marks1=sc.nextInt();
		System.out.println("Enter Class Test-2 Marks:");
		marks2=sc.nextInt();
	}
	void disp_marks()
	{
		System.out.println("Test-1 Marks:"+marks1);
		System.out.println("Test-2 Marks:"+marks2);
	}
}
```

```
class SingleInheritanceDemo
{
	public static void main(String args[])
	{
		Test t1=new Test();
		t1.get_stud_info();
		t1.get_marks();
		t1.disp_stud_info();
		t1.disp_marks();
	}
}
```

OUTPUT

```
Enter Student Roll No:
1010
Enter Student Name:
Dennis
Enter Class Test-1 Marks:
89
Enter Class Test-2 Marks:
67
Roll No:1010
Student Name:Dennis
Test-1 Marks:89
Test-2 Marks:67
```

Program: Multilevel Inheritance

```
import java.util.*;
class Student
{
	int rollno;
	String name;
	void get_stud_info()
	{
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter Student Roll No:");
		rollno=sc.nextInt();
		System.out.println("Enter Student Name:");
		name=sc.next();
	}
	void disp_stud_info()
	{
		System.out.println("Roll No:"+rollno);
		System.out.println("Student Name:"+name);
	}
}
class Test extends Student
{
	int marks1,marks2;
	void get_marks()
	{
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter Class Test-1 Marks:");
		marks1=sc.nextInt();
		System.out.println("Enter Class Test-2 Marks:");
		marks2=sc.nextInt();
	}
	void disp_marks()
	{
		System.out.println("Test-1 Marks:"+marks1);
		System.out.println("Test-2 Marks:"+marks2);
	}
}
class Result extends Test
{
```

```
        int total;
        void disp_total()
        {
                total=marks1+marks2;
                System.out.println("Total Marks:"+total);
        }
}
class MultilevelInheritanceDemo
{
        public static void main(String args[])
        {
                Result r1=new Result();
                r1.get_stud_info();
                r1.get_marks();
                r1.disp_stud_info();
                r1.disp_marks();
                r1.disp_total();
        }
}
```

OUTPUT

```
Enter Student Roll No:
1010
Enter Student Name:
Dennis
Enter Class Test-1 Marks:
99
Enter Class Test-2 Marks:
88
Roll No:1010
Student Name:Dennis
Test-1 Marks:99
Test-2 Marks:88
Total Marks:187
```

Program: Hierarchical Inheritance

```
import java.util.*;
class Student
{
	int rollno;
	String name;
	void get_stud_info()
	{
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter Student Roll No:");
		rollno=sc.nextInt();
		System.out.println("Enter Student Name:");
		name=sc.next();
	}
	void disp_stud_info()
	{
		System.out.println("Roll No:"+rollno);
		System.out.println("Student Name:"+name);
	}
}
class Test extends Student
{
	int marks1,marks2;
	void get_marks()
	{
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter Class Test-1 Marks:");
		marks1=sc.nextInt();
		System.out.println("Enter Class Test-2 Marks:");
		marks2=sc.nextInt();
	}
	void disp_marks()
	{
		System.out.println("Test-1 Marks:"+marks1);
		System.out.println("Test-2 Marks:"+marks2);
	}
}
class Sport extends Student
{
	int sport_wt;
	void get_sportwt()
```

```
	{
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter Sport Weightage:");
		sport_wt=sc.nextInt();
	}
	void disp_sportwt()
	{
		System.out.println("Sport Weightage:"+sport_wt);
	}
}

class HierarchicalInheritanceDemo
{
	public static void main(String args[])
	{
		Test t1=new Test();
		Sport s1=new Sport();
		System.out.println("****Test Class*****");
		t1.get_stud_info();
		t1.get_marks();
		t1.disp_stud_info();
		t1.disp_marks();
		System.out.println("****Sport Class*****");
		s1.get_stud_info();
		s1.get_sportwt();
		s1.disp_stud_info();
		s1.disp_sportwt();
	}
}
/*
****Test Class*****
Enter Student Roll No:
1010
Enter Student Name:
James
Enter Class Test-1 Marks:
99
Enter Class Test-2 Marks:
88
Roll No:1010
Student Name:James
Test-1 Marks:99
```

```
Test-2 Marks:88
****Sport Class*****
Enter Student Roll No:
1010
Enter Student Name:
James
Enter Sport Weightage:
9
Roll No:1010
Student Name:James
Sport Weightage:9
*/
```

Program: Hybrid Inheritance

```
import java.util.*;
class Student
{
	int rollno;
	String name;
	void get_stud_info()
	{
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter Student Roll No:");
		rollno=sc.nextInt();
		System.out.println("Enter Student Name:");
		name=sc.next();
	}
	void disp_stud_info()
	{
		System.out.println("Roll No:"+rollno);
		System.out.println("Student Name:"+name);
	}
}
class Test extends Student
{
	int marks1,marks2;
	void get_marks()
	{
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter Class Test-1 Marks:");
		marks1=sc.nextInt();
		System.out.println("Enter Class Test-2 Marks:");
		marks2=sc.nextInt();
	}
	void disp_marks()
	{
		System.out.println("Test-1 Marks:"+marks1);
		System.out.println("Test-2 Marks:"+marks2);
	}
}
class Result extends Test
{
```

```
        int total;
        void disp_totalMarks()
        {
                total=marks1+marks2;
                System.out.println("Total Marks:"+total);
        }
}
class Sport extends Student
{
        int sport_wt;
        void get_sportwt()
        {
                Scanner sc=new Scanner(System.in);
                System.out.println("Enter Sport Weightage:");
                sport_wt=sc.nextInt();
        }
        void disp_sportwt()
        {
                System.out.println("Sport Weightage:"+sport_wt);
        }
}

class HybridInheritanceDemo
{
        public static void main(String args[])
        {
                Result r1=new Result();
                Sport s1=new Sport();
                System.out.println("****Result Class*****");
                r1.get_stud_info();
                r1.get_marks();
                r1.disp_stud_info();
                r1.disp_marks();
                r1.disp_totalMarks();
                System.out.println("****Sport Class*****");
                s1.get_stud_info();
                s1.get_sportwt();
                s1.disp_stud_info();
                s1.disp_sportwt();
```

```
	}
}
```

```
****Result Class*****
Enter Student Roll No:
1010
Enter Student Name:
James
Enter Class Test-1 Marks:
88
Enter Class Test-2 Marks:
99
Roll No:1010
Student Name:James
Test-1 Marks:88
Test-2 Marks:99
Total Marks:187
****Sport Class*****
Enter Student Roll No:
1010
Enter Student Name:
James
Enter Sport Weightage:
7
Roll No:1010
Student Name:James
Sport Weightage:7
```

Program: Method Overriding

```
class Base
{
	void display()
	{
		System.out.println("display method of base class");
	}
}
class Derived extends Base
{
	void display()
	{
		super.display();
		System.out.println("display method of derived class");
	}
}
class MethodOverriding
{
	public static void main(String args[])
	{
		 Derived d1=new Derived();
		 d1.display();
	}
}
```

```
/*
OUTPUT
display method of base class
display method of derived class
*/
```

Program: To invoke base class constructor

```
class Base
{
	Base()
	{
		System.out.println("base class cosntructor called");
	}
}
class Derived extends Base
{
	Derived()
	{
		super();
		System.out.println("derived class constructor called");
	}
}
class InvokeBaseConstructor
{
	public static void main(String args[])
	{
		 Derived d1=new Derived();
	}
}
```

```
/*
base class cosntructor called
derived class constructor called
*/
```

Program: To invoke base class constructor

```
class Base
{
	Base(int x)
	{
		System.out.println("base class cosntructor called:"+x);
	}
}
class Derived extends Base
{
	Derived(int m,int n)
	{
		super(m);
		System.out.print
	}
}
class InvokeBaseConstructor1
{
	public static void main(String args[])
	{
		 Derived d1=new Derived(100,200);
	}
}
```

```
/*
base class cosntructor called:100
derived class constructor called:200
*/
```

Program: To avoid method overriding

```
class Base
{
	final void display()
	{
		System.out.println("display method of base class");
	}
}
class Derived extends Base
{
	void display()
	{
		System.out.println("display method of derived class");
	}
}
class AvoidMethodOverriding
{
	public static void main(String args[])
	{
		 Derived d1=new Derived();
		 d1.display();
	}
}
```

Program: To avoid Inheritance

```
final class Base
{
	void display()
	{
		System.out.println("display method of base class");
	}
}
class Derived extends Base
{
	void show()
	{
		System.out.println("show method of derived class");
	}
}
class AvoidInheritance
{
	public static void main(String args[])
	{
		Derived d1=new Derived();
		d1.display();
		d1.show();
	}
}
```